HPC User support and Onboarding

Solutions for HPC: Best Practices and Common Sense

Tuesday, August 4, 2026

SDSC
SAN DIEGO SUPERCOMPUTER CENTER

UC San Diego

Agenda

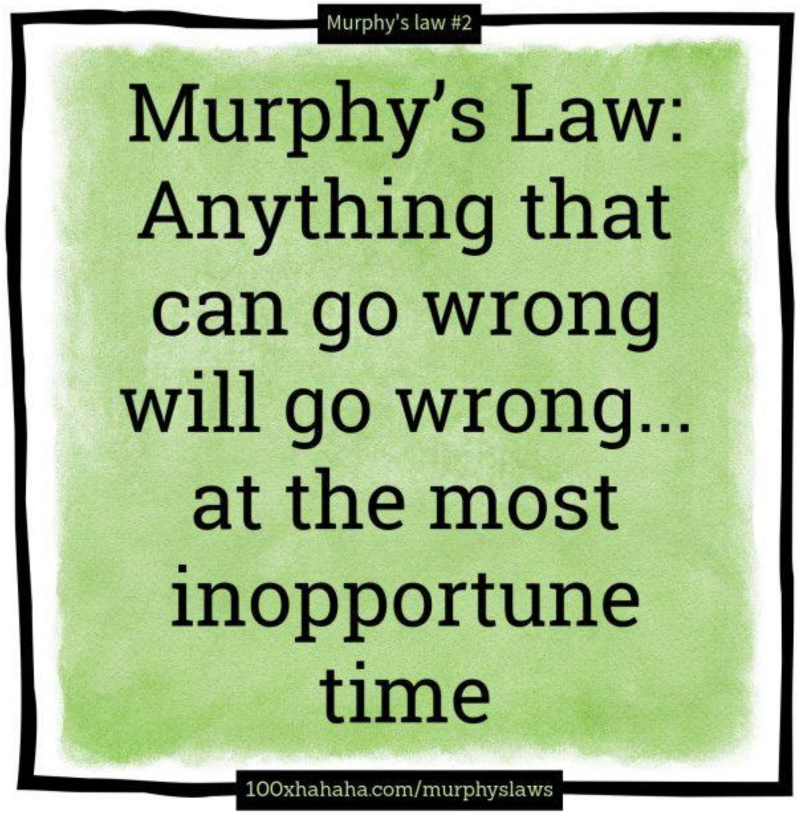
- HPC Ecosystem
- Available Help Resources
- Common Issues & solutions
  - Login, file systems, performance, batch scheduler, allocations
- Getting an allocation of your own
  - New NSF requirements
- Take Aways
- Resources

HPC Ecosystem

- You are not alone (HPC is a shared resource)
  - As a user (HPC ecosystem)
  - In getting support (available resources)
- What all is shared
  - Login Nodes
  - File systems
  - Compute nodes
  - System Administration, Security, Networking and System support teams
    - System policies

General Tips

- Plan Ahead
  - Don’t leave HPC tasks to the last minute
- Design for recovery
  - Back up your data
  - Use version control



What are the available Help options?

- What are the available resources(you are not alone)
  - User
  - Project/Colleagues
  - Web (Community forums, User guides, Git repositories)
  - Helpdesk

Best Practices: Getting help

- Understand your problem
- Engage with appropriate support tools
- Provide relevant and adequate information for helpdesk to reduce iterations
- Always be nice to the support desk! ☺

Before Asking for Help

- Check system status pages
- Review documentation
- Search existing FAQs and knowledge bases
- Examine job logs and error messages
- Verify account allocations and quotas
- Test with a small job

Types of HPC Support Available

- Training workshops and tutorials
- User forums and community channels
- Documentation and knowledge bases
- Help desk or ticketing system
- Office hours and consultation services
- Research computing consultants

Training and Educational Resources

- User onboarding materials
- Workshops and webinars
- Example job scripts
- Best practices guides
- Self-paced learning resources
  - https://www.sdsc.edu/education/on-demand-learning/index.html

How to Submit an Effective Support Request

- Include:
  - User ID/account name
  - Cluster/system name
  - Job ID(s)
  - Error messages
  - Commands or scripts used
  - Software versions/modules
  - Steps to reproduce the problem
  - Time the issue occurred
  - Example:"Job 123456 on Cluster X failed with a memory error after running for 2 hours. Attached are the job script and output logs."

HPC Support Etiquette and Expectations

- Recall you are not alone………
- Response times and service levels
- Prioritization of issues
- Following up on tickets
- Sharing resolutions with collaborators

Common Issues

- Login and Account problems
- Jobs/Policies/Schedulers
- Allocations/Job charging
- Applications/software
- Storage, Data storage, and Data Sharing
- Performance

Best Practices: Secure your credentials

- Passwords
    - Don’t reuse passwords
    - Longer is better
    - Don’t keep digital plaintext copies of passwords
    - Don’t Hard code password in files
    - Don’t share passwords
- Use password-manager program
- Use SSH keys, ssh-agent
- Multi-factor Authentication(MFA)

Common Issues: Login and account issues

- Access (first time)
  - Different resources have different access protocols, or mechanisms for access, even if its at the same site
    - Direct login
    - Portals
  - Password and username issues
    - Problem: Indicator message: Enter verification code
    - Solution: check username https://allocations.access-ci.org/profile (if username is not available for the resource then the account has not been created yet) Generally it can take 1 business day for accounts to be fully functional.
  - ssh keys
    - Problem: Indicator message: Enter password
    - Solution: resend ssh pub key, or use –i option to point to location of private key
  - MFA(2FA)

Common Issues: Resource Access/Logging In

- Unable to access system(returning user)
  - Cant log in
    - Problem: System Maintenance
    - Solutions:
      - Check specific error message
      - User News
      - Stay Subscribed to be notified
      - Check on User Portal -- https://support.access-ci.org/announcements
  - Hanging on login
    - Problem: Users overstimulating the file system
      - Solution: Support staff contact users to change behaviors
    - Problem: Activating Conda in your .bashrc
      - Solution: Do not initialize conda in .bashrc, but only when needed in run scripts

Common Issues: Job Submissions

- Submitting Jobs
  - Missing Allocation
  - Missing Software/License
  - Improper QOS
  - Job Scripts
    - Allocations
    - Queues/Partitions
    - Time Limits
    - OOM
    - New Software stack
      - Module spider
    - Expanse-client tool
    - Error message
      - sbatch: error: Project balance is not enough to run the job
      - sbatch: error: QOSMaxNodePerJobLimit
      - sbatch: error: Batch job submission failed: Job violates accounting/QOS policy (job submit limit, user's size and/or time limits)

```
login01 user_support]$ sbatch azton.sb
sbatch: error: Batch job submission failed: Invalid account or
account/partition combination specified
```

```
login01 user_support]$ sbatch azton.sb
sbatch: error: QOSMaxCpuPerJobLimit
sbatch: error: Batch job submission failed: Job violates accounting/QOS
policy (job submit limit, user's size and/or time limits)
```

Best Practices: jobs and job charging

- Check user guide for accounting policies
- Use system tools for most up to date accounting information
  - Scheduler tools: Slurm for individual job details
    - squeue
    - sacctmgr
    - scontrol
    - sinfo
    - sacct
  - Home grown tools for accounting information
    - Expanse-client

squeue

- Reports job status and reason codes
  - Queued Jobs (*ReqNodeNotAvail, Reserved for maintenance)

```
[nickel@login01 ~]$ squeue | more
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
          13574113   compute 80dgree_   yweng3 PD       0:00      2 (MaxMemPerLimit)
          12668967   compute 0-xtensi  kavousi PD       0:00      1 (MaxMemPerLimit)
          14756880   compute job001_p  amytsai PD       0:00     10 (Reservation)
          14800161   compute namd-com sasadian PD       0:00      6 (QOSMaxCpuPerUserLimit)
          14800218   compute namd-com sasadian PD       0:00      6 (QOSMaxCpuPerUserLimit)
          14789098   compute bl_8JHNp    uscms PD       0:00      1 (MaxJobsPerAccount)
```

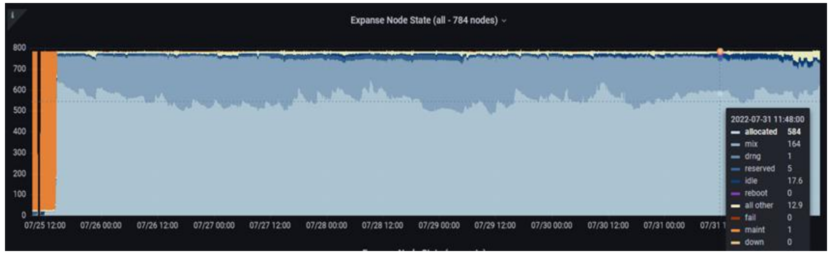
  - Running jobs

```
14813206   compute post0110 lpegolot  R   16:30:28      1 exp-9-35
14800090   compute namd-com sasadian  R   16:13:01      6 exp-2-29,exp-3-23,exp-4-33,exp-7-20,exp-9-[03,26]
14764467   compute V1WTReRU  aminkvh  R   16:08:56      1 exp-2-54
14773832   compute V4R1639Q  aminkvh  R   15:55:58      1 exp-8-14
14800092   compute namd-com sasadian  R   15:29:28      6 exp-4-29,exp-7-[07,39-40],exp-9-[28,41]
14812166   compute  scratch mlaskow2  R   15:53:59      1 exp-10-20
14812167   compute  scratch mlaskow2  R   15:39:34      1 exp-8-48
14800158   compute namd-com sasadian  R   15:17:18      6 exp-2-[26,50],exp-4-[52-53],exp-7-[42-43]
14812168   compute  scratch mlaskow2  R   15:20:01      1 exp-10-37
```

Common Issues: Job Submissions

- squeue – Common "reasons" for pending jobs
  - MaxMemPerLimit
    - Max. mem per Node = 243G
  - QOSMaxNodePerUserLimit
  - Priority
  - ReqNodeNotAvail, Unavailable nodes: exp-x-xx
- File system not available
  - We have added a slurm directive #SBATCH –constraint = "lustre" to indicate if your job is using the lustre file system. If this is provided, the scheduler will not launch the job on a node that is missing lustre.
- System Maintenance
  - https://support.access-ci.org/outages

sacctmgr



- View or modify Slurm account information
- sacctmgr show qos
  - Lists policies on queue/partitions, wait times, Time Limits
- User Guide

```
login01 ~]$ sacctmgr show qos format=name%20,MaxWall,MaxTRESPU%20,MaxJobsPU,MaxSubmitPU,MaxTRESPA%20,MaxJobsPA,MaxSubmitPA
                Name     MaxWall            MaxTRESPU MaxJobsPU MaxSubmitPU            MaxTRESPA MaxJobsPA MaxSubmitPA
-------------------- ----------- -------------------- --------- ----------- -------------------- --------- -----------
              normal  2-00:00:00     cpu=8192,node=64        32          64  cpu=16384,node=128        32          64
       shared-normal  2-00:00:00     cpu=8192,node=64      4096        4096  cpu=16384,node=128      4096        4096
 large-shared-normal  2-00:00:00                              2           4                             4           4
      preempt-normal  7-00:00:00                                        128     cpu=4096,node=32                  128
          gpu-normal  2-00:00:00 cpu=160,gres/gpu=16+         4           8   gres/gpu=32,node=8         8          16
   gpu-shared-normal  2-00:00:00 cpu=240,gres/gpu=24+        24          24 cpu=320,gres/gpu=32+        24          24
  gpu-preempt-normal  7-00:00:00   gres/gpu=24,node=6        12          16  gres/gpu=48,node=12        16          20
        debug-normal    01:00:00       cpu=256,node=2         2           2       cpu=256,node=2         2           2
    gpu-debug-normal    00:30:00     gres/gpu=8,node=2        2           2     gres/gpu=8,node=2         2           2
```

Common Issues: Queue and Time Limits

Partition Name	Max Walltime	Max Nodes/Job	Max Running Jobs	Max Running + Queued Jobs	Charge Factor	Notes
compute	48 hrs	32	32	64	1	Exclusive access to regular compute nodes; *limit applies per group*
ind-compute	48 hrs	32	32	64	1	Exclusive access to Industry compute nodes; *limit applies per group*
shared	48 hrs	1	4096	4096	1	Single-node jobs using fewer than 128 cores
ind-shared	48 hrs	1	32	64	1	Single-node Industry jobs using fewer than 128 cores
gpu	48 hrs	4	4	8 (32 Tres GPU)	1	Used for exclusive access to the GPU nodes
ind-gpu	48 hrs	4	4	8 (32 Tres GPU)	1	Exclusive access to the Industry GPU nodes
gpu-shared	48 hrs	1	24	24 (24 Tres GPU)	1	Single-node job using fewer than 4 GPUs
ind-gpu-shared	48 hrs	1	24	24 (24 Tres GPU)	1	Single-node job using fewer than 4 Industry GPUs
large-shared	48 hrs	1	1	4	1	Single-node jobs using large memory up to 2 TB (minimum memory required 256G)
debug	30 min	2	1	2	1	Priority access to shared nodes set aside for testing of jobs with short walltime and limited resources
gpu-debug	30 min	2	1	2	1	Priority access to gpu-shared nodes set aside for testing of jobs with short walltime and limited resources; *max two gpus per job*
preempt	7 days	32		128	.8	Non-refundable discounted jobs to run on free nodes that can be pre-empted by jobs submitted to any other queue
gpu-preempt	7 days	1		24 (24 Tres GPU)	.8	Non-refundable discounted jobs to run on unallocated nodes that can be pre-empted by higher priority queues

https://www.sdsc.edu/support/user_guides/expanse.html#running

scontrol

- View or modify Slurm configurations and state on currently queued or running jobs
- scontrol [OPTIONS...] [COMMAND...]
- Show individual job information
  - scontrol show job <<local_jobid>>

```
login01]$ scontrol show job 32658556
JobId=32658556 JobName=NGBW-JOB-BEAST2_XSEDE-43842C4A2D8C4B5085E71285ADC4D5DB
   UserId=testuser(505687) GroupId=use300(6099) MCS_label=N/A
   Priority=4509 Nice=0 Account=sds121 QOS=shared
   JobState=RUNNING Reason=None Dependency=(null)
   Requeue=1 Restarts=0 BatchFlag=1 Reboot=0 ExitCode=0:0
   RunTime=00:46:22 TimeLimit=03:00:00 TimeMin=N/A
   SubmitTime=2024-07-30T12:22:15 EligibleTime=2024-07-30T12:22:15
   AccrueTime=2024-07-30T12:22:15
   StartTime=2024-07-30T12:22:16 EndTime=2024-07-30T15:22:16 Deadline=N/A
   SuspendTime=None SecsPreSuspend=0 LastSchedEval=2024-07-30T12:22:16
Scheduler=Main
   Partition=shared AllocNode:Sid=login01:2295441
   ReqNodeList=(null) ExcNodeList=(null)
   NodeList=exp-16-53
   BatchHost=exp-16-53
   NumNodes=1 NumCPUs=1 NumTasks=1 CPUs/Task=1 ReqB:S:C:T=0:0:*:*
   ReqTRES=cpu=1,mem=1G,node=1,billing=3600
   AllocTRES=cpu=1,mem=1G,node=1,billing=3600
   Socks/Node=* NtasksPerN:B:S:C=1:0:*:* CoreSpec=*
   MinCPUsNode=1 MinMemoryNode=1G MinTmpDiskNode=0
   Features=(null) DelayBoot=00:00:00
   Reservation=cipres-shortjobs
   OverSubscribe=OK Contiguous=0 Licenses=cipres:1 Network=(null)
   Command=./_batch_command.run
   WorkDir=/expanse/projects//NGBW-JOB-BEAST2_XSEDE-
43842C4A2D8C4B5085E71285ADC4D5DB
   StdErr=/expanse/projects//NGBW-JOB-BEAST2_XSEDE-
43842C4A2D8C4B5085E71285ADC4D5DB/_scheduler_stderr.txt
   StdIn=/dev/null
   StdOut=/expanse/projects//NGBW-JOB-BEAST2_XSEDE-
43842C4A2D8C4B5085E71285ADC4D5DB/_scheduler_stdout.txt
   Power=
   MailUser=user@sdsc.edu
MailType=INVALID_DEPEND,BEGIN,END,FAIL,REQUEUE,STAGE_OUT
```

Sinfo: Why is my job not running? Queue, wait times, Time Limits

- View information about nodes and partitions

```
login01 ~]$ sinfo
PARTITION      AVAIL  TIMELIMIT  NODES  STATE NODELIST
compute           up 2-00:00:00      1  inval exp-1-15
compute           up 2-00:00:00      2 drain$ exp-13-[55-56]
compute           up 2-00:00:00      3 drain* exp-2-07,exp-5-50,exp-9-23
compute           up 2-00:00:00      7   drng exp-3-30,exp-4-17,exp-5-
[18,29,39,53],exp-7-45
compute           up 2-00:00:00      1  drain exp-4-55
compute           up 2-00:00:00     12   resv exp-2-[21-24],exp-14-50,exp-16-
[54-56],exp-17-[53-56]
compute           up 2-00:00:00    230    mix exp-1-[01-03,12-14,16,18,20,22-
26,28-39,41,43-44,47-48,51-54,56],
```

https://www.sdsc.edu/support/user_guides/expanse.html#running

sacct

- View accounting data for completed jobs and job steps
- Usage: sacct [OPTIONS...]

```
[nickel@login01 ~]$ sacct -j 28582088 -l
JobID        JobIDRaw        JobName  Partition  MaxVMSize  MaxVMSizeNode  MaxVMSizeTask  AveVMSize     MaxRSS MaxRSSNode MaxRSSTask
   AveRSS MaxPages MaxPagesNode   MaxPagesTask   AvePages     MinCPU MinCPUNode MinCPUTask     AveCPU   NTasks  AllocCPUS    Elapsed
     State ExitCode AveCPUFreq ReqCPUFreqMin ReqCPUFreqMax ReqCPUFreqGov     ReqMem ConsumedEnergy  MaxDiskRead MaxDiskReadNode MaxD
iskReadTask    AveDiskRead MaxDiskWrite MaxDiskWriteNode MaxDiskWriteTask   AveDiskWrite    ReqTRES  AllocTRES TRESUsageInAve TRESUsa
geInMax TRESUsageInMaxNode TRESUsageInMaxTask TRESUsageInMin TRESUsageInMinNode TRESUsageInMinTask TRESUsageInTot TRESUsageOutMax TRE
SUsageOutMaxNode TRESUsageOutMaxTask TRESUsageOutAve TRESUsageOutTot
```

- sacct -j 123456 format=jobid, partition%20
- For completed jobs it is encouraged to include where the job files are located including submit script, .err, and .out files

Common Issues: Accounting/Charging

- All systems charge differently
- ACCESS allocates in ACCESS Credits which can be converted to SUs (service Unit)
  - Each resource has a unique definition of an SU
  - Some are in core hours, some are in node hours, etc
  - Some charge for other components such as memory
- Visit the user guide for specifics
  - Check for opportunities to save
- Allocations are shared
- Charging is generally based on what is requested, not how resources are used
- Do test jobs to evaluate
  - Slurm commands to collect information
    - Sacct –u $USER
    - Sacct –j $JOBID

For Expanse Specifically…..

- Charging is based on what is requested, not how resources are used
- Charging is based on the Maximum of memory and CPU core fraction
- Minimum charge for any job is 1SU

Example for CPU

Max [3600 * #CPU cores, 1800 * #Mem in GB] / 3600 * (wallclock time in secs/3600)

- 1 CPU and less then 2GB of memory are charged 1 CPU Service Unit(1SU = 1 core/hour).
- 1 GPU and up to 10 CPUs and 92 GB of memory are charged 1 GPU Service Unit (SU)/hour. This job will be charged 1 GPU SU/hour.
- The minimum charge for any job is 1 SU.
- 1 Expanse SUs = 1 ACCESS Credit
- 1 Expanse GPU SU = 54 Expanse SUs (for conversion)
  - https://allocations.access-ci.org/exchange_calculator

Accounting

- SDSC uses slurm as a scheduling tool for cluster management and accounting
- Different resources use different home grown tools to help users evaluate their usage.
  - Expanse-client tool(SDSC)
  - Projects (PSC)
  - taccinfo (TACC)

```
login01 ~]$ expanse-client user train112 -p

 Resource  expanse

 NAME      STATE  PROJECT  TG PROJECT     USED  AVAILABLE  USED BY PROJECT
-----------------------------------------------------------------------
 train112  allow  gue998   TG-CIE960001S   337    200000        114621
```

- ACCESS Portal updated at various intervals

Common Issues: Software

- Software Installs
  - Modules
  - Compile (Build your own)
  - Containers
- Most HPC system use module to manage their software stacks
- Login nodes, compute nodes are different
  - Compiling codes need to happen on the nodes on which they will run
  - Modules need to be loaded on the nodes on which applications will run
- Software managed by the resource provider has usually been tested and fine tuned for the specific resources

Best Practices: Software

- Review user guide for tools available
- Use system installed applications when available
- Use containers to manage out of date software

Environment Modules

- Environment Modules help manage software incompatibilities, versioning, and dependencies
- Environment Modules provide for dynamic modification of your shell environment
- Module commands set, change, and/or delete environment variables
- Modules manage software versions
- Module manage dependencies by loading or unloading other modules
    - Check for dependencies with module spider <module_name>
- Module list - lists all the currently loaded modules

Navigating with Modules

Command	Description
`module list`	Currently Loaded Modules
module avail	List of available software modules based on your current module path
module spider	List all available software and versions on the system
Module spider <application name>	List available application specific modules and module details, including versions and dependencies
module load [module file]	Load module(s) or specify unresolved dependencies
module show [module file]	Display information about loaded modules including changes, dependencies, versions and paths
module unload [module file]	Unloads a specified module form the environment
module purge	Unloads all the loaded modules
module reset	Reset modules to default settings

Common Issues: File systems

- Cant write to home
- Sharing
- File system hanging
- Backups

What is in a file system

- Common File systems and their utilization
  - Home – Usually limited in space
  - Scratch - Large space, limited persistence, no backup
  - Node local scratch(SSD)- good performance, only available during job
  - Persistent storage – good performance, limited persistence
  - Archival- Slow, backup
- Types of file systems
  - Lustre
  - NFS
  - GPFS
  - Ceph

File management

- Review file permissions ls –la
- Review usage
  - lfs
  - du
- Review user group
  - Groups username, id username
- Controls: Permissions granularity levels
  - (Attribute), User, Group, Other
  - Read(4), Write(2), Execute(1)
  - Default 755 (User(read, write, execute):Group(read, execute): Other(read, execute)
  - Use chmod, chown commands to modify ownership and permissions
  - X is capitalized so that group execute permission is given to any files that are executables (don't use lower case because that would change all files to have execute permissions).

Have a backup plan

- Copy critical data of system regularly
- Version Control: Git
- Checksum data transfers to ensure no corruption
- ACCESS to HPC system and file retention is usually limited. Transferring data takes time
- Data on the system is the users responsibility
- Plan ahead for data transfers
- Convert many small files into a single archive file before transfer

Best Practices: Backups

- Review user guide for file systems availability and usage guides
- Backups should be done at regular intervals that make sense to your project
  - Frequency
  - Ensure backups are made on “good” versions
  - Perhaps retain a few versions just in case
- Don’t back up everything
  - Clean up unnecessary files
  - Backup files not easily reproduced or replaced
    - source code, scripts, config files and large output files
- Backups should be on a different resource
- Ensure credentials would not allow hackers to get onto the external resource
- Test the backup plan with the restore process

Common Issue: system performance

- GPU
- CPU
- Login Nodes
- Lustre

Monitoring Resource Utilization

- Tools
  - top, htop, atop : display Linux processes
  - mpstat : display processors related statistic
  - sar : display system activity report
  - free: display memory statics
  - ps: display active processes
  - nvidia-smi: display gpu activity and statistic
- seff (slurm efficiency) after job is complete

Allocations

- SDSC allocates resources via two methods: peer review process or for purchase via three different programs: NSF allocated resources(ACCESS-CI), Discretionary time(HPC@UC), and TSCC & Industrial program.
  - ACCESS-CI: (Advanced Cyberinfrastructure Coordination Ecosystem: Services & Support) (https://access-ci.org/)
  - NAIRR: (National AI research resource)(https://nairrpilot.org/)
  - HPC@UC: Available to UC folks to gain quick access to limited compute and storage resources with goal to graduate to ACCESS proposal
  - HPC@MSI: Available to folks at MSI to gain quick access to limited compute and storage resources with goal to graduate to ACCESS proposal
  - TSCC(Triton Shared Compute Cluster: Purchase resources on our Condo/Hotel cluster
  - Industrial Partners: Purchase SUs on our Expanse Industry Rack

ACCESS: Types of ACCESS Allocations

- **Trial allocation: 100 GPU and/or 1000 CPU hours**
  - **Submit request to: consult@sdsc.edu**
- **Explore Allocations**
  - Small, fast-turnaround requests
  - Ideal for testing and new users
- **Discover Allocations**
  - Startup projects and educational activities
- **Accelerate Allocations**
  - Medium-sized research projects
- **Maximize Allocations**
  - Large-scale computational research projects
- **Apply for allocations through ACCESS**
  - **https://allocations.access-ci.org/prepare-requests-overview**

Allocation	Credit Threshold
Explore ACCESS	400,000
Discover ACCESS	1,500,000
Accelerate ACCESS	3,000,000
Maximize ACCESS	Not awarded in credits.

After Receiving an Allocation

- Request local accounts at resource providers
- Complete required training
- Learn resource-specific policies
- Monitor usage and balances
- Request extensions or renewals when needed

HPC@UC

- Request HPC@UC at: https://www.sdsc.edu/support/hpc_uc_apply-exp.html
- Up to 500K core-hours of computing, associated data storage, and access to SDSC expertise to assist their research team.
- Awards are active for one year. NO supplements, renewals or Extensions
- Applicants must not have an active ACCESS award
- Developed to support onboarding to ACCESS and larger, formal allocation requests
- SDSC staff will assist in developing these allocation applications.
- Applications are reviewed on an ongoing basis. Applicants will be notified within 10 business days of the review decision.
- https://www.sdsc.edu/collaborate/hpc_at_uc.html

Writing a Strong Allocation Request

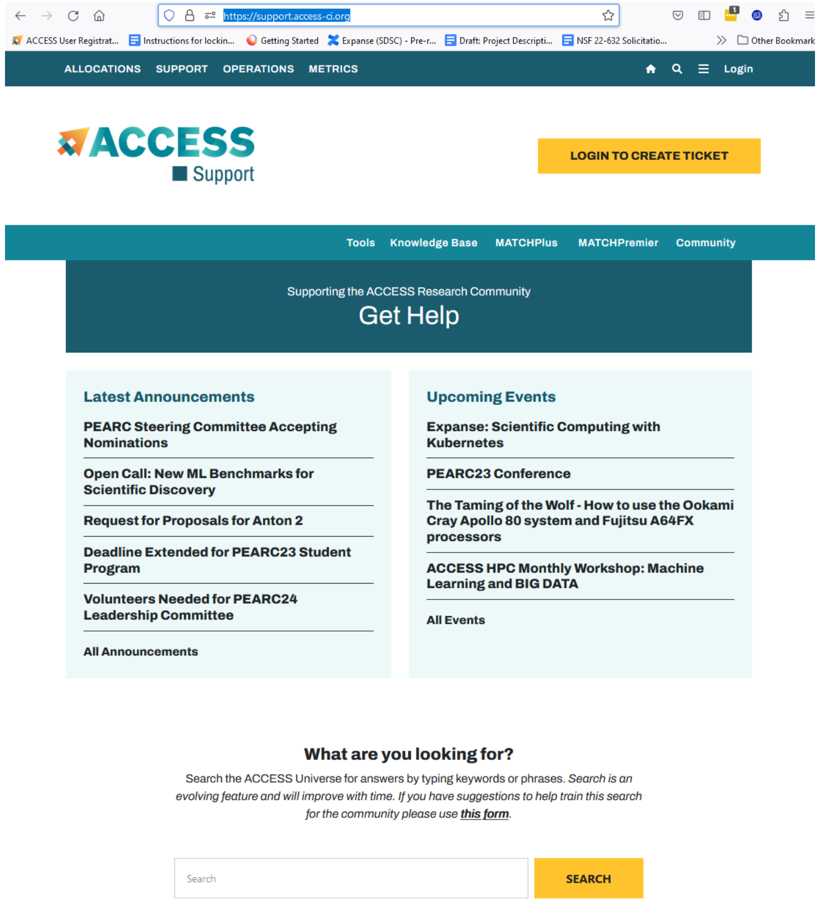
- **Resource Justification**
- Explain:
  - Number of CPU/GPU hours needed
  - Storage requirements
  - Software requirements
  - Scaling tests (if available)
- **Common Mistakes**
  - Overestimating resources
  - Vague project descriptions
  - Missing computational methodology
  - Insufficient justification

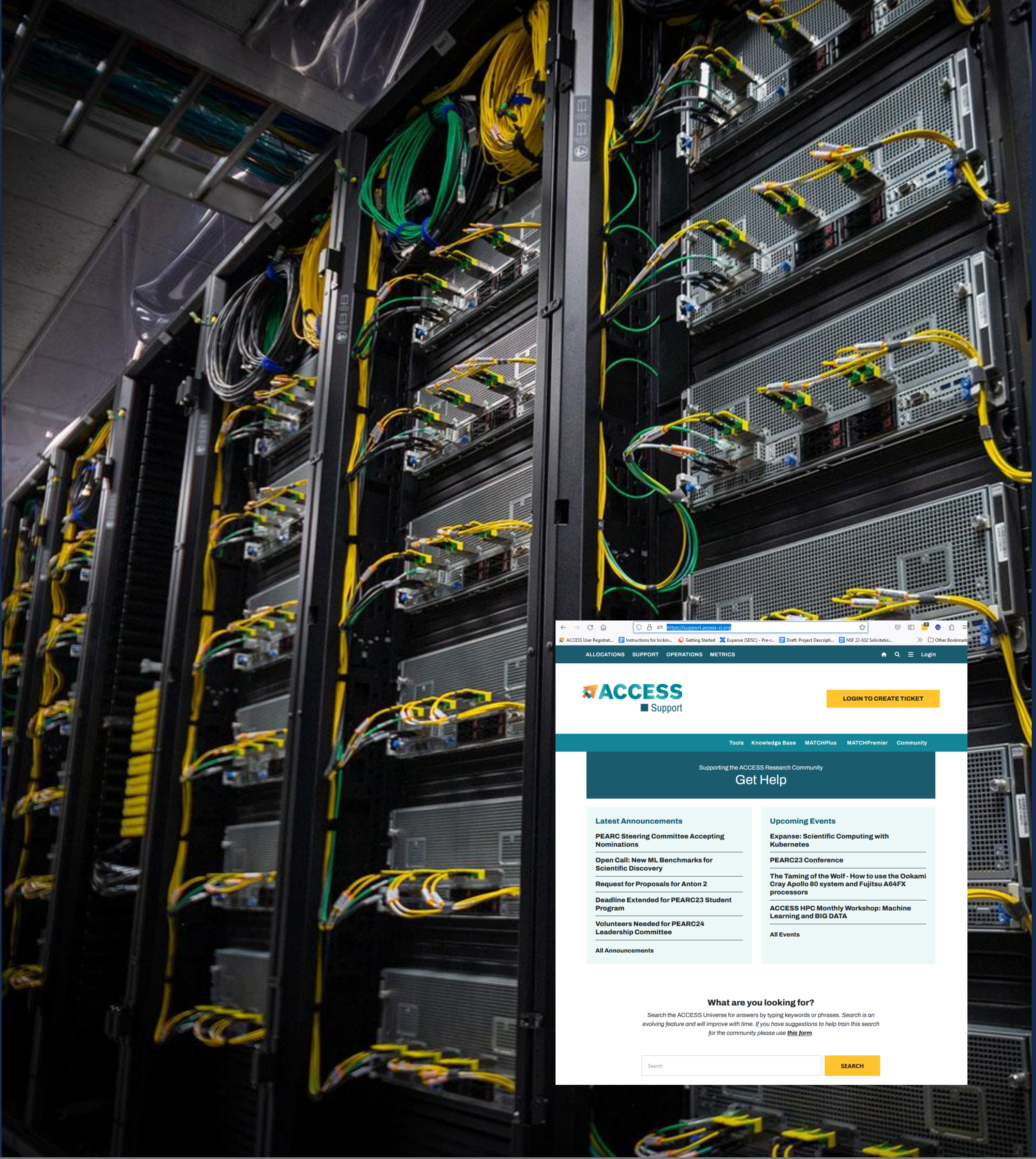
Key Takeaways

- Use available documentation first
- Collect relevant information before contacting support
- Provide complete details in support requests
- Take advantage of training and consultation services
- Engage support early for complex computational projects

Resources

- ACCESS Allocations Portal
  - https://allocations.access-ci.org
- ACCESS User Support:
  - https://support.access-ci.org
- ACCESS Documentation:
  - https://docs.access-ci.org
- SDSC HPC User Support
  - consult@sdsc.edu
- SDSC Documentation
  - https://sdsc.edu/





Questions

- SDSC Support
  - consult@sdsc.edu
- ACCESS Allocations Portal: ACCESS Allocations Portal
- https://allocations.access-ci.org
- ACCESS User Support: ACCESS Support
- https://support.access-ci.org
- ACCESS Documentation: ACCESS Documentation
- https://docs.access-ci.org